

Kamal Jutt

Senior Data Engineer | ETL & MLOps Specialist | Cloud Data Platform Expert

✉ kamal.jutt.dev@gmail.com ☎ (253) 893-4306 📍 New York City, NY, 10018

Professional Summary

Results-driven Lead Data Engineer with over 10+ years of experience designing, building, and optimizing cloud-native data platforms across AWS, Azure, and GCP. Expert in ETL/ELT pipelines, real-time data streaming, and big data solutions using Spark, Kafka, and Snowflake. Proven record of reducing data processing time by 65%, cutting infrastructure cost by 30%, and enabling real-time analytics for 2M+ messages per second. Skilled in MLOps, CI/CD, and data governance frameworks, ensuring high-quality, secure, and scalable data delivery. Adept at collaborating with cross-functional teams, leading engineers, and translating business needs into actionable data solutions. Recognized for driving innovation, improving system reliability, and accelerating data-driven decision-making across enterprises.

Skills

Programming & Scripting

Python, SQL, Scala, Java, Shell Scripting, REST APIs,

Analytics & BI Tools

Tableau, Power BI, Looker, Qlik, Superset, Mode Analytics, Excel (Advanced), Data Visualization Best Practices

Machine Learning & Advanced Analytics

(Supportive) Feature Engineering, ML Data Preparation, Model Deployment Support (MLflow, SageMaker, Vertex AI), A/B Testing Data Pipelines, Real-time Analytics, Data for AI/LLM Workloads

Cloud Platforms & Services

AWS (S3, Redshift, Glue, EMR, Lambda, Athena, Kinesis), Azure (Data Lake, Synapse Analytics, Data Factory, Databricks), GCP (BigQuery, Dataflow, Dataproc, Pub/Sub), Snowflake, Databricks, Terraform Cloud

Databases & Storage

Relational Databases (PostgreSQL, MySQL, Oracle, SQL Server), NoSQL (MongoDB, Cassandra, DynamoDB, Redis, Couchbase), Columnar Databases (Redshift, BigQuery, Vertica, Greenplum), Graph Databases (Neo4j, Amazon Neptune)

Snowflake

Designed and optimized Snowflake data warehouses and ETL workflows.

Data Automation

Data Governance & Security

Data Quality Frameworks, Data Cataloging, Metadata Management, Master Data Management (MDM), Data Lineage, GDPR, HIPAA, SOC 2, Role-Based Access Control (RBAC), Data Masking & Encryption

Data Engineering & ETL

ETL/ELT Development, Data Pipeline Orchestration, Data Modeling (OLTP & OLAP), Data Warehousing, Data Integration, Real-time & Batch Processing, Workflow Scheduling (Airflow, Luigi, Oozie, Azkaban)

DevOps, CI/CD & Infrastructure

CI/CD Pipelines (Jenkins, GitLab CI, GitHub Actions, Azure DevOps), Docker, Kubernetes, Terraform, Ansible, Infrastructure as Code (IaC), Serverless Architectures, Monitoring & Logging (Datadog, Prometheus, Grafana, Splunk, ELK Stack)

Big Data & Distributed Systems

Apache Spark, Hadoop Ecosystem (Hive, Pig, HDFS, MapReduce), Kafka, Flink, Presto, Delta Lake, Druid, Storm

Professional Experience

Senior Data Engineer, Secoda

12/2022 – Present

- Led design and implementation of cloud-native data platforms (AWS, Azure) supporting real-time analytics and predictive insights, improving data accessibility by 60%.
- Built scalable Kafka-based pipelines processing 2M+ messages per second, reducing latency by 40% and improving data reliability.
- Architected ETL/ELT workflows in Spark and Snowflake, cutting data processing time by 65% and boosting pipeline efficiency.
- Drove data governance & compliance (GDPR, HIPAA) frameworks, ensuring 100% audit-ready datasets across business domains.
- Partnered with Data Science teams to deliver ML-ready datasets, improving model training accuracy by 70% and reducing deployment cycles by 30%.
- Directed CI/CD & DevOps integration (Jenkins, Datadog, Terraform), reducing deployment time by 50% and enhancing observability.
- Spearheaded legacy-to-cloud migration, achieving a 25% cost reduction and improving infrastructure scalability.
- Mentored 5+ engineers, fostering best practices in data reliability, observability, and automation.
- Executed role-based access data masking, ensuring secure access to sensitive data and meeting SOC 2 compliance requirements.
- Created self-service analytics frameworks, enabling business users to generate insights independently and reducing BI team workload by 35%.
- Introduced predictive alerting for pipeline failures using Spark and Datadog, reducing downtime and manual interventions by 50%.

Big Data Engineer, Rill Data

08/2018 – 11/2022

- Engineered high-volume big data pipelines using Apache Spark, Kafka, and Airflow, processing 10B+ records daily across distributed systems.
- Optimized Snowflake and Redshift data warehouses, reducing BI query latency by 55% and improving reporting performance.
- Automated ETL workflows and orchestration frameworks, reducing manual operations by 70% and boosting reliability.
- Modernized data infrastructure by migrating on-prem Hadoop clusters to AWS EMR and Azure Data Lake, cutting infrastructure cost by 30%.
- Built data quality & anomaly detection frameworks, increasing data accuracy and integrity by 80%.
- Collaborated cross-functionally with product & BI teams to design business-aligned data models, improving decision-making speed by 45%.
- Contributed to DevOps and CI/CD adoption, ensuring seamless integration, version control, and monitoring across data platforms.
- Implemented event-driven architecture with Kafka for real-time data processing, enabling near-instant business insights.
- Designed partitioning and indexing strategies for Snowflake, improving query efficiency and reducing compute costs by 20%.

- Developed and maintained ETL pipelines for structured/unstructured data across Hadoop and SQL ecosystems, improving data delivery SLAs by 40%.
- Built SQL-based data models, stored procedures, and views, enhancing analytics efficiency and supporting enterprise-wide BI dashboards.
- Implemented data validation and cleansing frameworks, increasing data accuracy by 90% and reducing manual QA effort by 50%.
- Designed data visualization dashboards using Tableau and Power BI, delivering actionable insights to business teams.
- Contributed to Hadoop adoption (Hive, Pig) for batch processing, enabling scalable data warehousing and analytics.
- Partnered with business teams to translate analytics requirements into technical data architecture solutions.

Projects

Real-Time Data Streaming Platform, Kafka + Spark + Snowflake

- Architected and deployed a real-time data ingestion system handling 2M+ Kafka messages per second across distributed clusters.
- Integrated Spark Structured Streaming with Snowflake for real-time analytics and alerting, cutting latency by 45%.
- Achieved 99.9% message delivery reliability and reduced downstream lag through partition optimization and schema evolution.
- Tech Stack: Kafka, Spark, Snowflake, AWS Kinesis, Airflow, Terraform, Datadog

Cloud Data Lake Modernization, AWS & Databricks

- Managed migration from on-prem Hadoop to AWS-based Databricks platform, improving data scalability and reducing storage cost by 35%.
- Built data ingestion frameworks using AWS Glue and Delta Lake, automating schema tracking and lineage.
- Planned cost-optimized storage tiering strategy using S3 and Redshift Spectrum for hybrid workloads.
- Tech Stack: Databricks, AWS S3, Redshift, Glue, Delta Lake, Python, Terraform

Machine Learning Data Pipeline Automation

- Developed ML data preparation pipelines supporting model training and feature store population in Snowflake and Databricks.
- Enabled automate retraining triggers and versioned feature storage, improving model refresh efficiency by 60%.
- Supported MLOps deployment via MLflow and SageMaker, integrating CI/CD pipelines for reproducible experiments.
- Tech Stack: Snowflake, Databricks, MLflow, AWS SageMaker, Airflow, Python

Education

Bachelor of Science in Computer Science